of the gauge coupling, Eq. (23.5.20) gives for simple gauge groups

$$\int d^4x\, \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma} = 64\pi^2 \nu / g^2 \,, \tag{27.3.20}$$

where $\nu = 0, \pm 1, \pm 2, \ldots$ is an integer, the winding number, which characterizes the topological class of the gauge field configuration. Thus for instantons of winding number ν, the Lagrangian density $\mathscr{L}_\theta$ contributes a phase to path integrals, given by

$$\left[\exp\left(i \int d^4x\, \mathscr{L}_\theta\right)\right]_\nu = \exp(i\nu\theta) \,, \tag{27.3.21}$$

so the effects of $\mathscr{L}_\theta$ are periodic in θ, with period 2π.

It is often convenient to absorb a factor g into the gauge field, so that the structure constants do not depend on g, and instead the Lagrangian density for the gauge field is multiplied by an over-all factor $1/g^2$. In this notation, the complete Lagrangian density for the gauge field may be written in terms of the rescaled gauge fields and structure constants as

$$\mathscr{L}_{\text{gauge}} + \mathscr{L}_\theta = -\text{Re}\left[\frac{\tau}{8\pi i} \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{AL\alpha} W_{AL\beta}\right]_{\mathscr{F}} \,, \tag{27.3.22}$$

where τ is the complex coupling parameter

$$\tau \equiv \frac{4\pi i}{g^2} + \frac{\theta}{2\pi} \,. \tag{27.3.23}$$

According to Eq. (23.5.19), the contribution of instantons of winding number ν to path integrals is suppressed by a factor $\exp(-8\pi^2|\nu|/g^2)$, which together with the factor (27.3.21) yields an over-all factor

$$\exp\left[i\nu\theta - \frac{8\pi^2|\nu|}{g^2}\right] = \begin{cases} \exp(2\pi i \nu\tau) & \nu \geq 0 \\ \exp(2\pi i \nu \tau^*) & \nu \leq 0 \end{cases} \,. \tag{27.3.24}$$

27.4 Renormalizable Gauge Theories with Chiral Superfields

We will now put together the pieces assembled in the previous three sections, to construct the most general renormalizable action for chiral superfields interacting with general gauge fields. Adding the terms (27.1.27), (27.2.7), and (27.3.1) and the superpotential terms in Eq. (26.4.5) gives the Lagrangian density

$$\mathscr{L} = \frac{1}{2}\left[\Phi^\dagger \exp\left(-2\sum_A t_A V_A\right)\Phi\right]_D - \frac{1}{2}\text{Re}\sum_A \left(W_{AL}^{\mathrm{T}} \epsilon W_{AL}\right)_{\mathscr{F}}$$
$$- \frac{g^2\theta}{16\pi^2}\sum_A \text{Im}\left(W_{AL}^{\mathrm{T}} \epsilon W_{AL}\right)_{\mathscr{F}} - \sum_A \xi_A [V_A]_D + 2\,\text{Re}\,[f]_{\mathscr{F}}$$